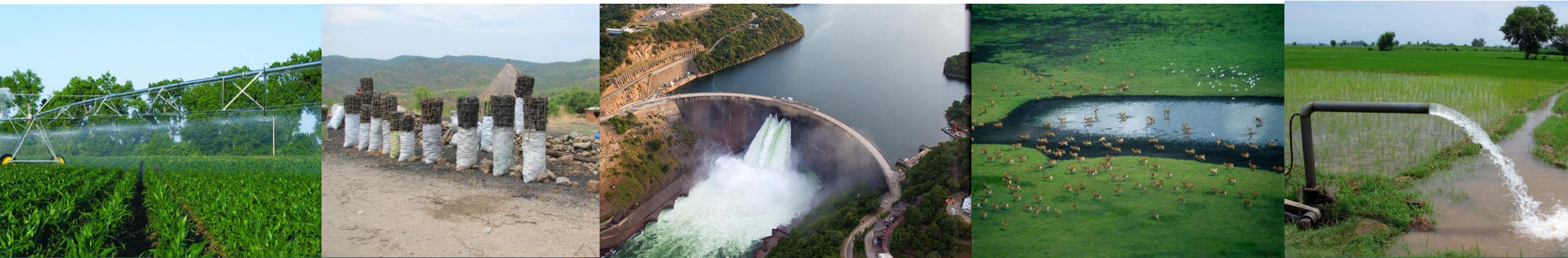


Exercise 9: Changing Meteo Data

Peter Burek, Carla Catania, Silvia Artuso
International Institute for Applied Systems Analysis
Water Security Group



Changing Meteo Data

1. What is netCDF
2. Display netCDF files
3. Meteorological data in CWatM



Changing Meteo Data

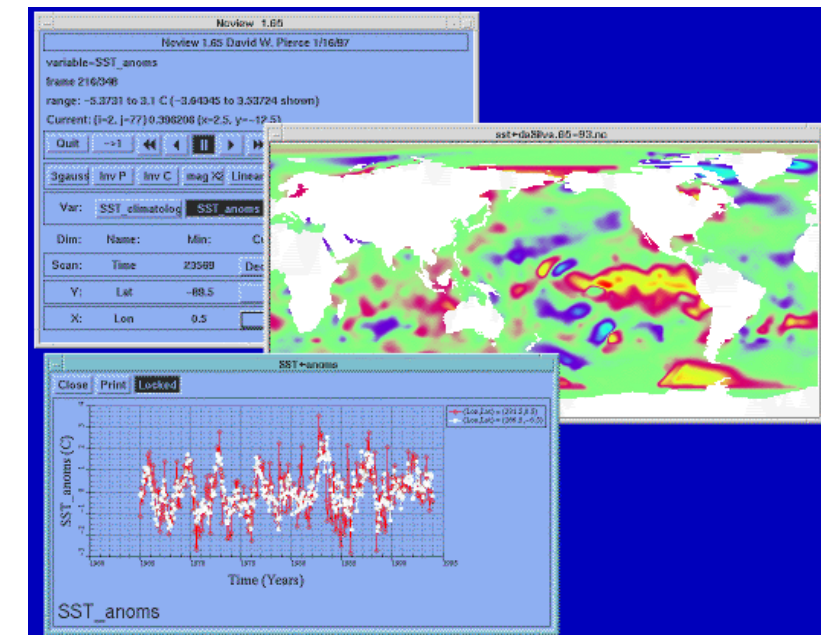
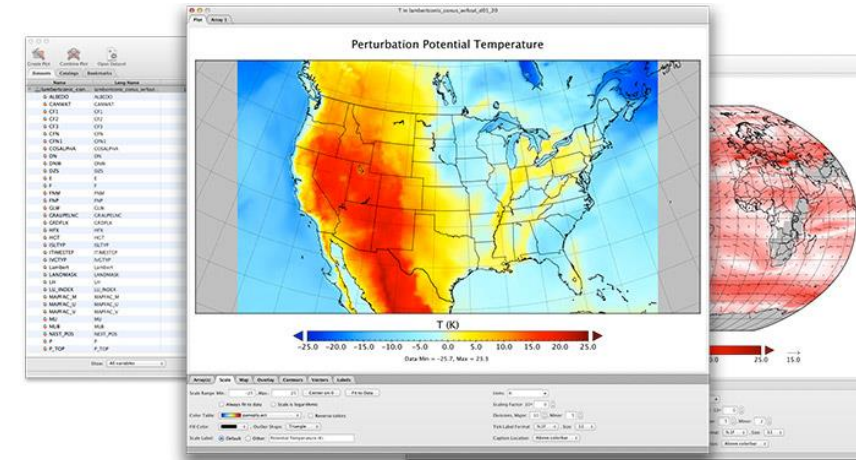
0. What you need

- A Geoinformation System e.g. QGIS, ArcGIS
- A way to display netCDF files easily e.g.:

PANOPLY on Windows

<https://www.giss.nasa.gov/tools/panoply/>

ncview on Linux:



Changing Meteo Data

What is netCDF

- **NetCDF (Network Common Data Form)** is a data format like .tif, .jpg
- **NetCDF** is a set of libraries and self-describing, machine-independent data formats that support the creation, access, and sharing of array-oriented scientific data
- Mainly used by the meteorological community

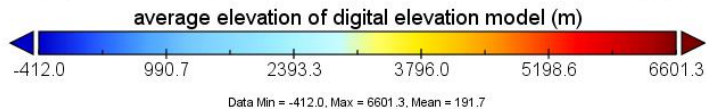
Advantage:

- netCDF is self-describing, portable, flexible
- netCDF is public domain, well documented and used by a growing number of organizations
- High efficiency in reading and processing netCDF files (it is fast)
- A netCDF file can contain timeseries of spatial data in a compressed way

Disadvantage:

- Not so easy to handle

Digital elevation (source:SRTM)



```
input5min/landsurface/topo/dem.nc {  
  
ital_elevation_model";  
  
84\","DATUM["D_WGS_1984\","SPHEROID["WGS_1984\","637813  
  
18";  
  
":
```

Changing input data

Clipping data with CDO (Linux)

CDO (Climate Data Operators)

<https://code.mpimet.mpg.de/projects/cdo/>

CDO is a collection of command line Operators to manipulate and analyse netCDF data.

There are more than 600 operators available

Metainformation e.g.

- dimension (here **with** time)
- unit
- description

```
cdo -f nc4 -z zip sellonlatbox,8,30,41,51 income.nc out.nc
```

Clipping dataset to Danube extent

Changing input data

Meteo dataset in CWatM

- Take a look at:
<https://cwatm.iiasa.ac.at/data.html>
- Take a look at
./yourbasin/meteo/emo-1

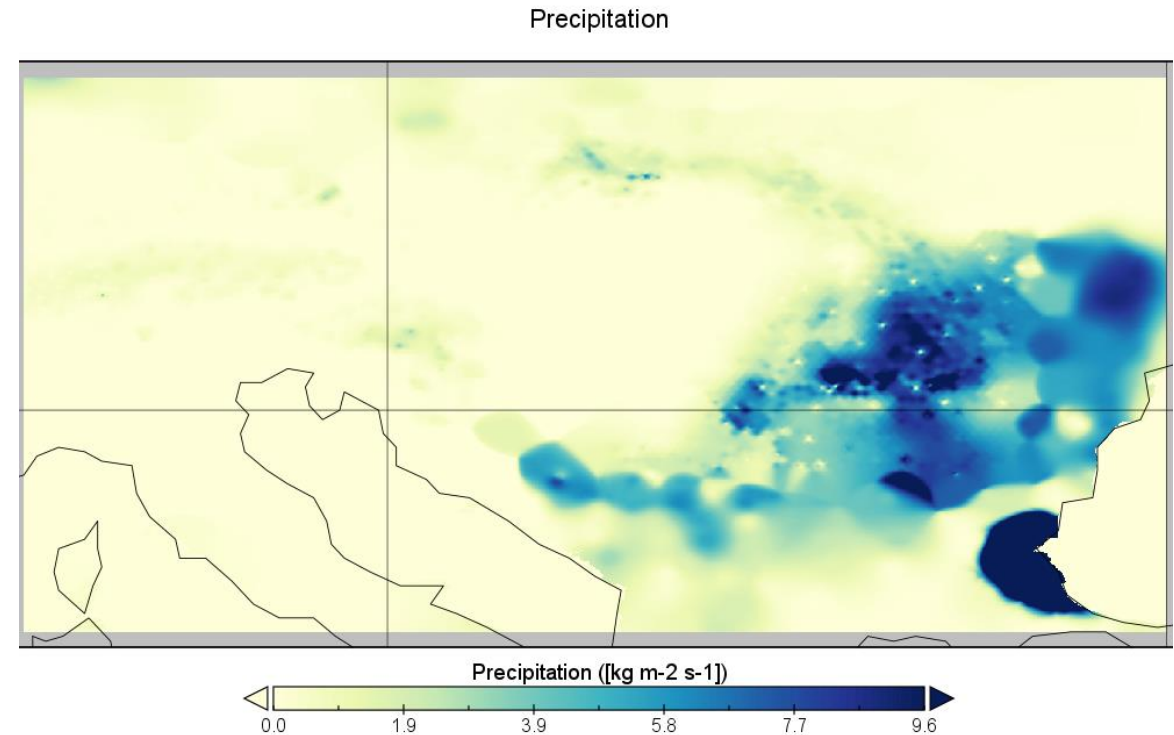
ETRef_morava.nc:	Potential evaporation (reference crop) as a result of Penman-Monteith calculation of evaporation [unit: m]
EWRef_morava.nc:	Potential evaporation (water) as a result of Penman-Monteith calculation of evaporation [unit: m]
pr_EMO-1_1990_2022:	Precipitation [unit: mm]
tas_EMO-1_1990_2022:	Average temperature [unit: °C]

Changing input data

Meteo dataset in CWatM

The full dataset to calculate potential evaporation with Penman-Monteith equations as daily NetCDFs

Pr:	precipitation [mm or m or kg m-2 s-1]
Tas, tasmin, taxmax:	average, minimum, maximum temperature [$^{\circ}\text{C}$ or K]
Wind:	Windspeed [m/s]
RG:	Incoming Radiation [W m^{-2}]
or:	
Rlds:	Downward longwave radiation [W m^{-2}]
Rlds:	Downward shortwave radiation [W m^{-2}]
Huss:	Specific humidity [kg kg^{-1}]
or:	
Hurs:	Relative humidity [%]
PS:	Surface pressure [pa]
or	
PD:	Vapor pressure [hPa]



Changing input data

How to go on:

- 1) Download data: e.g. Eobs: https://surfobs.climate.copernicus.eu/dataaccess/access_eobs.php#datafiles
- 2) Clip to Danube: `cdo -f nc4 -z zip sellonlatbox,8,30,41,51 rr_ens_mean_0.1deg_reg_v29.0e.nc rr_danube.nc` 3) Check and maybe 3)
- 3) Convert units
 - for radiation, windspeed, humidity, air pressure the units in CWatM are fixed – you have to convert before
 - for temperature you can either use °C or K (there is a flag in settingsfile)
 - for precipitation it can be any unit (there is a conversion factor in settingsfile)

Clipping information – map extent

Danube:	lon: 8-30	lat: 41-51
Morava:	lon: 15-18.5	lat: 48-50.5
Sava:	lon: 13-22	lat: 42-47
Tisa:	lon: 19-27	lat: 45-50

Changing input data

Changes in the settings file

- If Temperature is in °C change option

- if precipitation is [mm]:

precipitation_coversion = 0.001

- if precipitation is [kg m⁻² s⁻¹]:

precipitation_coversion = 86.4

Part of the settings file

which deals with meteodata

```
[METEO]
# if meteo maps have the same scale than all the other maps e.g. 0.5 deg
metemapssamescale = True
downscale_wordlim_avg = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/tas_monthmean.nc
downscale_wordlim_tmin = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/tasmin_monthmean.nc
downscale_wordlim_tmax = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/tasmax_monthmean.nc
downscale_wordlim_prec = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/pr_monthlysum.nc

# if usemeteodownscaling = True (default if not set) meteo maps will be downscaled
# if usemeteodownscaling = False: metemaps will be use as they come e.g. as 0.5 deg
usemeteodownscaling = False

# precipitation [kg m-2 s-1]
PrecipitationMaps = $(FILE_PATHS:PathMeteo)/pr_*
# average daily temperature [K]
TavgMaps = $(FILE_PATHS:PathMeteo)/tas_*

# -----
# This is used if calc_evaporation = False

# daily reference evaporation (free water)
EOMaps = $(FILE_PATHS:PathMeteo)/EWRref_morava.nc
# daily reference evapotranspiration (crop)
ETMaps = $(FILE_PATHS:PathMeteo)/ETref_morava.nc
# this is true, if ET maps have the same resolution as pr maps
ETsamePr = True
# -----
# from kg m-2s-1 to m : 86.4
#precipitation_coversion = 86.4

# from MM to m : 0.001
precipitation_coversion = 0.001
evaporation_coversion = 1.0
```

```
[OPTIONS]
#-----
# OPTION - to switch on/off
#-----

# if temperature is stored in Kelvin instead Celsius
TemperatureInKelvin = False

# from MM to m : 0.001
precipitation_coversion = 0.001
```

```
[EVAPORATION]
# This is used if calc_evaporation = True
# use albedo maps
albedo = False
albedoMaps = $(FILE_PATHS:PathMaps)/landsurface/albedo/albedo.nc

# if not albedo maps use fixed albedo
# Albedo of bare soil surface (Supit et. al.)
AlbedoSoil = 0.15
# Albedo of water surface (Supit et. al.)
AlbedoWater = 0.05
# Albedo of vegetation canopy (FAO,1998)
AlbedoCanopy = 0.23

only_radiation = True
dem = $(FILE_PATHS:PathMaps)/landsurface/topo/demavg.nc
latitude = $(FILE_PATHS:PathMaps)/landsurface/topo/lat.nc

# use CO2 correction from Yang et al. 2019 -> CO2 concentration
co2conc = $(FILE_PATHS:PathMaps)/RCP_CO2/ssp370.nc4
# use specific humidity (TRUE) Qair, or relative humidity (FALSE) - rhs
useHuss = False

# radiation maps [KJ m-2 day-1]
RGDMaps = $(FILE_PATHS:PathMeteo)/rg_*
# vapor pressure maps [hPa]
EActMaps = $(FILE_PATHS:PathMeteo)/pd_*

# map stacks Temperature [K]
TminMaps = $(FILE_PATHS:PathMeteo)/tasmin_*
TmaxMaps = $(FILE_PATHS:PathMeteo)/tasmax_*
# Instantaneous surface pressure[Pa]
##PSurfMaps = $(FILE_PATHS:PathMeteo)/ps_*
# 2 m instantaneous specific humidity[kg /kg] (Qair) or relative humidity [%] (rhs)
##RhsMaps = $(FILE_PATHS:PathMeteo)/hurs_*
# wind speed maps at 10m [m/s]
WindMaps = $(FILE_PATHS:PathMeteo)/ws_*
```

Changing input data

Automatic downscaling

CWatM Danube runs on 1 arcmin. It needs also meteo data on 1arcmin:

If meteo data are in 1 arcmin (e.g. EMO), no downscaling

If meteo data in e.g. 6 arcmin (e.g. Eobs),

automatic downscaling is used to produce 1 arcmin

Automatic Downscaling is using a reference year of high resolution meteo data for each month.

Low resolution is downscaled to high resolution data by using the Spatial pattern of the monthly reference high res data.

```
[METEO]

# if meteo maps have the same scale than all the other maps e.g. 0.5 deg
metemapssamescale = True
downscale_wordclim_tavg = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/tas_monthmean.
downscale_wordclim_tmin = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/tasmin_monthme
downscale_wordclim_tmax = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/tasmax_monthme
downscale_wordclim_prec = $(FILE_PATHS:PathMaps)/meteo/monthly_emol/pr_monthlysum.

# if usemeteodownscaling = True (default if not set) meteo maps will be downscaled
# if usemeteodownscaling = False: meteomaps will be use as they come e.g. as 0.5 d
usemeteodownscaling = False

# precipitation [kg m-2 s-1]
PrecipitationMaps = $(FILE_PATHS:PathMeteo)/pr_*
# average daily temperature [K]
TavgMaps = $(FILE_PATHS:PathMeteo)/tas_*

# -----
# This is used if calc_evaporation = False

# daily reference evaporation (free water)
EOMaps = $(FILE_PATHS:PathMeteo)/EWRef_morava.nc
# daily reference evapotranspiration (crop)
ETMaps = $(FILE_PATHS:PathMeteo)/ETRef_morava.nc
# this is true, if ET maps have the same resolution as pr maps
ETsamePr = True
# -----
# from kg m-2s-1 to m : 86.4
#precipitation_coversion = 86.4

# from MM to m : 0.001
precipitation_coversion = 0.001

evaporation_coversion = 1.0
```

Changing input data

Additional information:

- please take care of the right units and put on the right option in the settingsfile – using [mm] data and a conversion of 86.4 leads to flooding.
- Meteo data can come in a single file, but also split into year, decades
- Meteo data can be upside down in netCDF (data with low lat stored on top CWatM recognize this, and turns it automatically
- EMO data has only 1 radiation file, ISIMIP, MSWX have 2
for emo please put in settings [Meteo] only_radiation = True

 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1921_1930.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1931_1940.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1941_1950.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1951_1960.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1961_1970.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1971_1980.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1981_1990.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_1991_2000.nc
 gfdl-esm4_r1i1p1f1_w5e5_historical_pr_global_daily_2001_2010.nc